TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Passage Diagram and Production Calculations of Winding, Warping and Sizing Machines

The three preparatory machines that turn spinning packages into a sized weaver's beam: the passage of yarn through each machine, the parts to identify on the spot, and the production calculation each machine is judged by — worked through in full so you can repeat them with the lab machines' own figures.

TE-2104 · WEAVING TECHNOLOGY SESSIONAL · STUDY 1 · LEVEL 2 TERM 1

1. Object and method of the study

Object: to draw and understand the **passage diagram** — the exact route of the yarn through the machine — of the winding machine, the warping machine and the sizing machine, and to carry out the standard production calculation for each from observed speeds, counts and efficiencies.

Method: at each machine, stand at the feed side and walk the yarn's route to delivery, naming every guide, tensioner and roller it touches (Sections 2–4 give the routes). Then take the machine's figures — delivery speed, yarn count, efficiency — and work the production calculation in the keybox. A passage diagram you cannot draw from memory is a machine you do not yet understand; the examination of this study is built on exactly that.

> **THE COUNT RELATION USED IN EVERY CALCULATION BELOW**
>
> For cotton count Ne, one pound of yarn measures 840 × Ne yards = 768.1 × Ne metres. So: **mass (lb) = length (m) ÷ (768.1 × Ne)**, and kg = lb × 0.4536. Every production figure in this sheet is length first, then converted by this one relation.

2. The winding machine

2.1 Parts to identify along the passage

- **Supply package** — ring-frame cop (for warp winding) or cone/cheese, on the creel peg at the base of the winding head.

- **Yarn guide and tension device** — disc (or gate) tensioner that sets the winding tension; note the washer combination fitted.
- **Yarn clearer** — mechanical slub catcher (a calibrated slit) or electronic clearer; find the setting and state what faults it removes (thick places, slubs) and where the cut ends go (the knotter/splicer that rejoins them).
- **Traverse guide / drum** — on a drum winder the grooved drum both drives the package by surface contact and traverses the yarn; on a precision winder a separate traverse guide does this job.
- **Delivery package** — the cone or cheese being built; note its size and the winding density impression (a package wound too soft or too hard both cause trouble at warping).
- **Stop motion** — the head stops when the yarn breaks or the supply runs out; find the feeler that detects it.

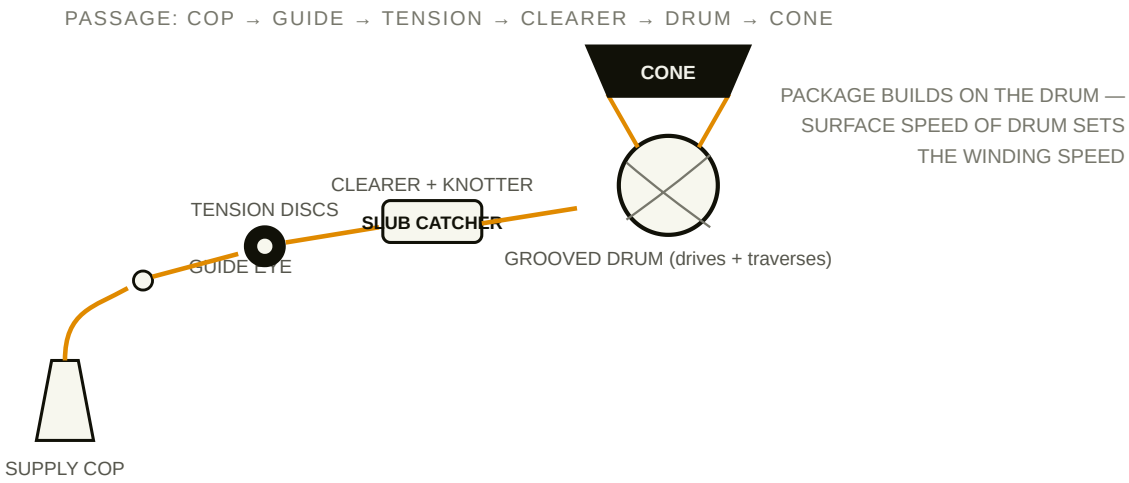


Figure 1 — Passage of yarn through one winding head (drum winder). Winding is not only repackaging: the clearer cuts out the faults the ring frame made, and the knotter or splicer rejoins the yarn invisibly.

WORKED CALCULATION — WINDING PRODUCTION

Winding speed 1,000 m/min; yarn 20 Ne; efficiency 80%; shift 8 h.

Length wound per drum = 1,000 × 60 × 8 × 0.80 = 384,000 m.

Mass = 384,000 ÷ (768.1 × 20) = 384,000 ÷ 15,362 ≈ **25.0 lb ≈ 11.3 kg per drum per shift**.

Repeat with the lab machine's speed and count: production rises in direct proportion to speed, and falls in proportion to count number (finer yarn = fewer kg for the same metres).

3. The warping machine

3.1 Parts to identify along the passage

- **Creel** — rows of cones, one per warp end; count the creel capacity and the ends actually crept for the beam in preparation. Find the tension device at each peg and the **stop motion** (electric or photo-electric) that stops the machine and lights a lamp when an end breaks.
- **Guide/expansion reed** — the adjustable reed that spreads the sheet of ends to the required warp width.
- **Measuring roller** — the roller over which the sheet passes to the beam, driving the length counter.
- **Warping beam (beam warping)** — the flanged beam on which the full sheet is wound in parallel; *or* the **sectional warping drum** — the cone-faced drum on which bands (sections) are wound one at a time and later beamed off together. Identify which type the lab machine is.
- **Beaming arrangement** — on a sectional machine, how the sections are transferred to the weaver's beam under even tension.

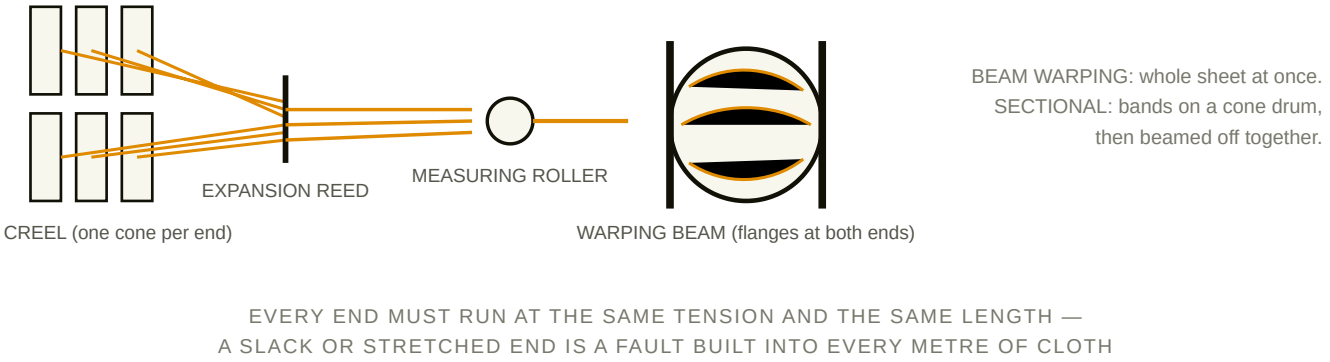


Figure 2 — Passage through a beam warping machine. Hundreds of ends leave the creel as one sheet, are spread to width by the reed, measured, and wound as a parallel band onto the warping beam.

WORKED CALCULATION — WARPING PRODUCTION

Beam carries 400 ends; required warp length on the beam 5,000 m; yarn 20 Ne; warping speed 600 m/min.

Time to wind one beam at full speed = 5,000 ÷ 600 ≈ **8.3 min** (creel stops and end-mending will make the real time longer — that difference is the warping efficiency).

Total yarn on the beam = 400 × 5,000 = 2,000,000 m.

Mass = 2,000,000 ÷ (768.1 × 20) = 2,000,000 ÷ 15,362 ≈ **130.2 lb ≈ 59.1 kg of warp per beam**.

4. The sizing machine

4.1 Parts to identify along the passage

- **Beam creel** — the series of warping beams (often 8–16) whose sheets combine into one; count the beams and multiply by ends per beam to get the total ends in the warp.
- **Size box** — find the immersion roller that carries the sheet under the size liquor, the size circulation/heating, and the pair of **squeeze rollers** that set the wet pick-up. Note the size recipe board if posted (adhesive — starch/PVA/CMC; softeners; antiseptic) and the liquor temperature.
- **Drying cylinders** — the steam-heated cans over which the wet sheet is dried; count them and find the temperature indication.
- **Headstock** — after drying: the **dividing rods (lease rods)** that split the sheet back into its beam layers (the size film glues ends together; splitting must separate them without breaking), the expansion reed, and the **weaver's beam** being wound under controlled tension.
- **Marking motion** — the device that marks the cloth length (cut lengths) on the warp as it is beamed.

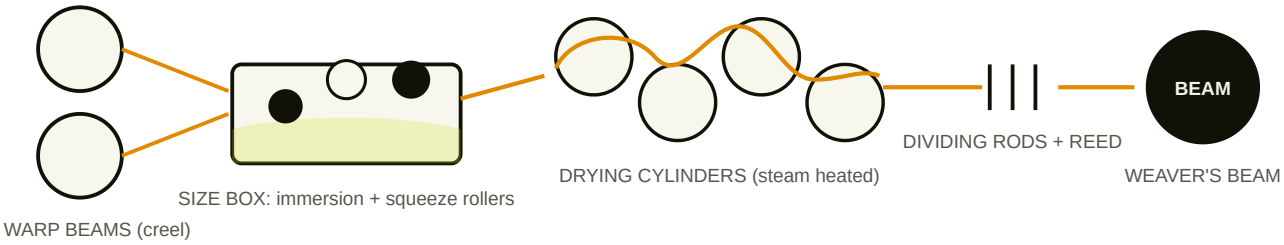


Figure 3 — Passage through the sizing machine (slasher). The combined sheet from all the warping beams is immersed, squeezed, dried over the cans, split into its layers at the dividing rods, and wound as the weaver's beam.

WORKED CALCULATION — SIZE ADD-ON AND BEAMING TIME

Add-on: dry warp in the set = 100 kg; required size add-on 12%. Dry size to deposit = 0.12 × 100 = **12 kg**. If the size liquor holds 10% solids, the wet pick-up needed = 12 ÷ 0.10 = **120 kg of liquor per 100 kg of warp (120% wet pick-up)**; the drying cylinders must then evaporate about 120 − 12 = 108 kg of water. Add-on is always quoted on the weight of the *unsized* dry warp.
Time: beam length 10,000 m at a sizing speed of 40 m/min = 10,000 ÷ 40 = **250 min ≈ 4.2 h** to beam one set (before stoppages).

5. From the sized beam to cloth — a loom production check

The three machines exist to feed the loom, so this study's calculations close with the loom's own figure, which Study 2 observes in operation:

WORKED CALCULATION — LOOM PRODUCTION

Loom speed 180 picks/min; cloth 60 picks/inch; efficiency 85%.

Picks inserted per hour = 180 × 60 = 10,800.

Cloth length per hour at full speed = 10,800 ÷ 60 picks/inch = 180 inches = 4.57 m.

At 85% efficiency: 4.57 × 0.85 ≈ **3.9 m of cloth per loom-hour**.

Note the chain: every stoppage the preparatory machines build in — a slack end from warping, a badly sized section — is paid for here, in loom stops.

MACHINE	TAKES IN	DELIVERS	JUDGED BY
Winding	Ring cops / small packages	Large cleared cones/cheeses	m/min per head; kg per drum per shift; faults cleared
Warping	Cones, one per end	Warping beam (or sections) of exact length and even tension	m/min; ends × length = total metres; beam mass
Sizing	Set of warping beams	Sized weaver's beam at set add-on	Add-on %; wet pick-up; m/min and beams per shift

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Draw the passage of yarn through a winding head from memory, and state the job of each element it touches.
2. Why is a clearer fitted on the winder when the yarn has already passed inspection at the ring frame?
3. A winder runs at 1,000 m/min on 20 Ne at 80% efficiency. Show that one drum produces about 25 lb in an 8-hour shift.
4. A warping beam holds 400 ends of 5,000 m of 20 Ne. Find the mass of warp on the beam.
5. What is the difference between beam warping and sectional warping, and when is sectional warping chosen?
6. Define size add-on. A warp of 100 kg (unsized) is to carry 12% add-on from a liquor of 10% solids: find the wet pick-up and the water the dryers must remove.
7. A loom makes 180 picks/min weaving cloth of 60 picks/inch at 85% efficiency. Find the cloth production per hour.

Answer key

1. Cop → guide eye → tension discs → clearer (slub catcher) with knotter/splicer → grooved drum → cone. Guide directs the yarn; tension discs set a steady winding tension; the clearer cuts out thick faults and the knotter rejoins; the drum drives the package by surface contact and its groove traverses the yarn to build the cone.
2. Because the ring frame inevitably spins in faults — slubs, thick and thin places, neps, double ends. Winding is the first point where yarn runs singly at high speed past a sensing point, so it is the economical place to remove them: every fault left in becomes a warp break or a cloth defect later.
3. Length = 1,000 × 60 × 8 × 0.80 = 384,000 m. Mass = 384,000 ÷ (768.1 × 20) = 384,000 ÷ 15,362 ≈ **25.0 lb**.
4. Total length = 400 × 5,000 = 2,000,000 m. Mass = 2,000,000 ÷ 15,362 ≈ **130.2 lb (≈ 59.1 kg)**.
5. Beam warping winds the full sheet of ends directly onto a flanged beam in one operation — fast, for large orders of one warp. Sectional warping winds the ends in narrow bands side by side on a cone-faced drum, then beams all bands off together; it is chosen for smaller lots, patterned/coloured warps and short runs, where creeling the full width for beam warping is not justified.
6. Add-on = dry size deposited as a % of the dry unsized warp weight. Size needed = 12 kg. Wet pick-up = 12 ÷ 0.10 = **120 kg liquor per 100 kg warp**. Water to evaporate = 120 − 12 = **108 kg**.
7. Picks per hour = 180 × 60 = 10,800; length = 10,800 ÷ 60 = 180 in = 4.57 m; at 85%: ≈ **3.9 m/h**.
